

Gender Differences in Political News Coverage: A Semantic and Emotional Analysis

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Abstract

This proposal examines how political news coverage differs based on the gender of the politician being covered, using computational text analysis methods. I gathered a balanced dataset of news articles (150 articles per politician, across 7 female and 7 male politicians) from LexisNexis news archives. I apply **Structural Topic Modeling (STM)** to uncover latent topics and assess differences in topic prevalence by politician gender. I conduct an **emotion analysis** using the NRC Emotion Lexicon to compare the frequency of emotion-laden words, normalized per 1,000 words, between articles about female versus male politicians. Additionally, I perform a **word embedding analysis** (Word2Vec) to detect semantic shifts in word usage between the two sets of articles. The preliminary results suggest notable differences: domestic topics (e.g., those related to personal life or specific policy areas) are more prominent in coverage of female politicians, while others international affairs (e.g., tariffs and war) appear more in male-politician coverage. Emotional tone also varies, with articles about male politicians containing slightly higher frequencies of negative emotion words. Word embedding comparisons reveal that some words (e.g., “woman,” “age,” “tension”) carry different contextual meanings in articles about female versus male politicians. These findings highlight subtle but meaningful gender-based disparities in media portrayal of politicians. I discuss the implications of these differences, methodological limitations (such as the focus on specific public figures and topics), and propose directions for future research to further unpack gender biases in political media.

Background and Motivation

Gender bias in media coverage of politicians has long been a topic of concern in political communication research. Studies have found that female politicians often receive coverage that differs in focus and tone compared to their male counterparts¹. For instance, news articles may emphasize personal life, family roles, or “feminine” qualities of female leaders more than they do for men. Conversely, male politicians might receive more coverage on policy stances or professional qualifications. These differences can reinforce stereotypes – portraying women as less serious or less

¹ Van der Pas, D., & Aaldering, L. (2020). Gender differences in political media coverage: A meta-analysis. *Journal of Communication*, 70(1), 114–143.

qualified – and may affect public perceptions of candidates. As women remain underrepresented in political office, understanding media portrayal is crucial. Biased coverage could discourage women from running for office or influence voter attitudes by perpetuating gendered frames (e.g. focusing on a woman candidate’s emotions or family instead of her policies).

In the current project, I aim to systematically analyze language used in political news to uncover gender-based differences in both content (topics and word usage) and tone (emotional language). Prior content analyses have often been manual or limited to smaller samples psypost.org. Our approach leverages computational methods on a larger scale to detect patterns that might be missed by human coding. By combining topic modeling, lexicon-based sentiment analysis, and word embedding techniques, I provide a multifaceted view of the differences in how media describe female versus male politicians. This study is motivated by the need for empirical evidence on subtle biases in news text that could cumulatively contribute to the “paper ceiling” female politicians face in media visibility and portrayal academic.oup.com. The findings will not only contribute to academic discussions on gender and media but also inform journalists and readers about potential unconscious biases in political reporting.

Data Collection

To investigate these questions, I compiled a comprehensive corpus of political news articles focusing on a selected set of politicians. I chose 14 prominent U.S. political figures – 7 female and 7 male – ensuring a balance in gender. The female politicians in our sample include Kamala Harris, Nancy Pelosi, Amy Klobuchar, Nikki Haley, Elizabeth Warren, Gretchen Whitmer, and Elise Stefanik. The male politicians include Joe Biden, Donald Trump, Bernie Sanders, Barack Obama, Ron DeSantis, Kevin McCarthy, and J.D. Vance. These individuals were selected for their prominence and to cover a range of offices (executive, legislative) and political affiliations, providing a broad view of media coverage.

For each of the 14 politicians, I retrieved approximately 150 news articles from the LexisNexis archives. LexisNexis is a large news database that aggregates articles from thousands of publications, offering deep archives of news content. The search strategy involved using each politician’s name as a keyword, filtering for news articles (excluding press releases or opinion pieces when possible) to ensure the content reflects journalistic reporting. Each retrieved article was labeled with metadata indicating the politician it is about and that politician’s gender. This metadata labeling allows us to later group and compare articles by the gender category. I also recorded other metadata such as the news source, publication date, and headline for reference and potential

future covariate analysis.

After initial collection, I performed **name-based filtering** to ensure relevance. Articles where the politician's name appeared only incidentally (for example, a long list of names in passing) were discarded. I kept articles primarily about or significantly featuring the target individual. This yielded a curated dataset of political news texts – 2,100 articles in total (150 per person $\times$ 14 persons). The corpus is balanced by design: half of the articles concern female politicians and half concern male politicians, which is important for making fair comparisons.

Text Preprocessing

Before analysis, all articles underwent thorough text preprocessing. I converted the text to lowercase and removed punctuation, numbers, and other non-word characters to clean the data. Standard stopwords (common words like “the”, “and”, “of”) were removed to focus on meaningful content words. In addition, I implemented a **custom stopwords** list to remove each politician's name and common variants (e.g., “Biden”, “Joe”, “Biden's”) from the articles. Removing names is critical because names could otherwise dominate topics or skew sentiment – for example, “Trump” appearing frequently doesn't carry substantive topical meaning by itself. By excluding names, I ensure the analysis focuses on what is being said *about* the person rather than the repetition of their name.

I also verified that no large text sections (e.g., copyright notices or repetitive boilerplate) are present, cleaning those if found. After these steps, I tokenized the text (split into words) and, for certain analyses, stemmed or lemmatized words to their base form (e.g., “running” $\rightarrow$ “run”). However, for interpretability, the topic modeling was performed on the original (unstemmed) words. I maintained an index of each article's total word count as well, which was later used for normalization (e.g., computing emotion word frequencies per 1,000 words). The end result of preprocessing was a clean document-term matrix for the corpus and ready-to-use text data for subsequent modeling.

Structural Topic Modeling (STM)

To uncover latent themes in the news coverage, I employed a **Structural Topic Model (STM)**, which is a form of topic modeling that can include document-level metadata (covariates). In our case, the key covariate is the politician's gender. The STM approach allows us to discover topics – groups of words that frequently appear together – and to examine how the prevalence of these topics differs in articles about female versus male politicians. I set the number of topics to 10, balancing detail with interpretability. The model was fitted using the `stm` package in R, with gender included

as a binary covariate influencing topic prevalence.

Each of the 10 discovered topics was interpreted and given a label based on its most representative words and a reading of example articles. The topics ranged across various political news themes. For example, I identified topics related to elections and campaigns, policy issues, economic and financial news, and even a topic about media/news content itself. The figure below shows the expected proportion of each topic in the overall corpus, giving a sense of the major themes present:

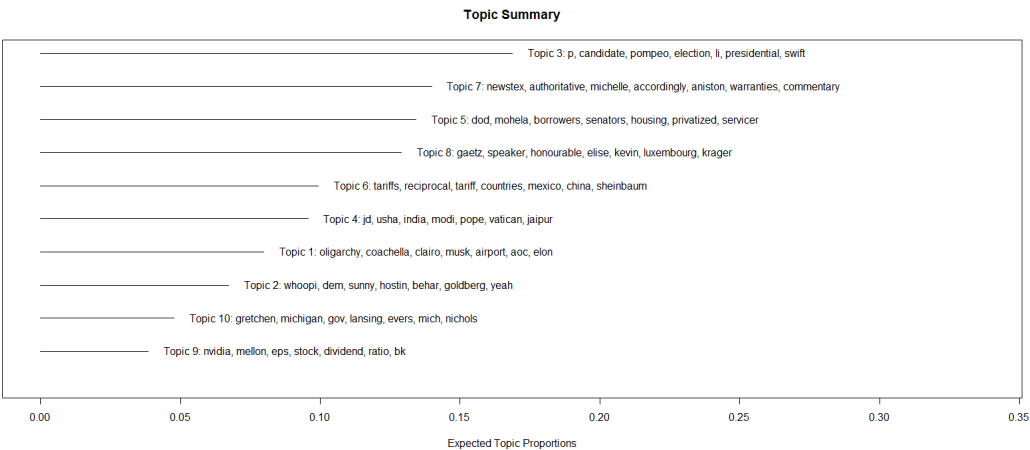


Figure 1: *STM Result (10-topic STM).*

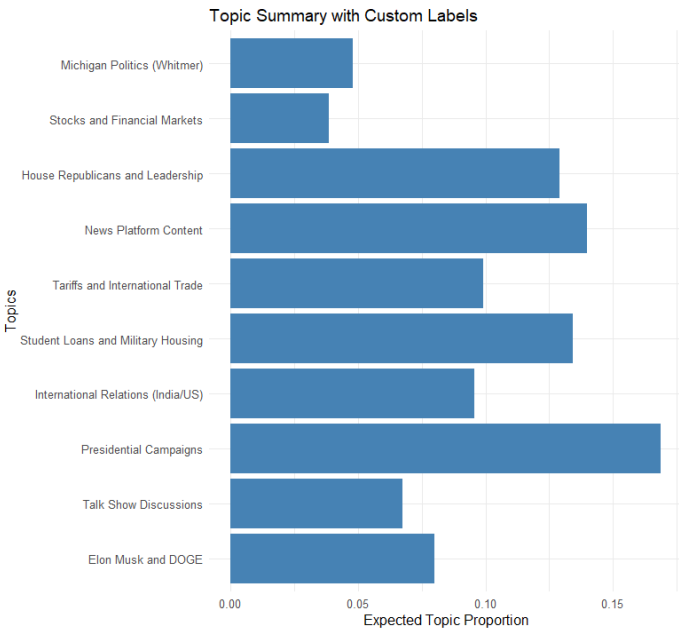


Figure 2: *Summary of discovered topics (10-topic STM)².*

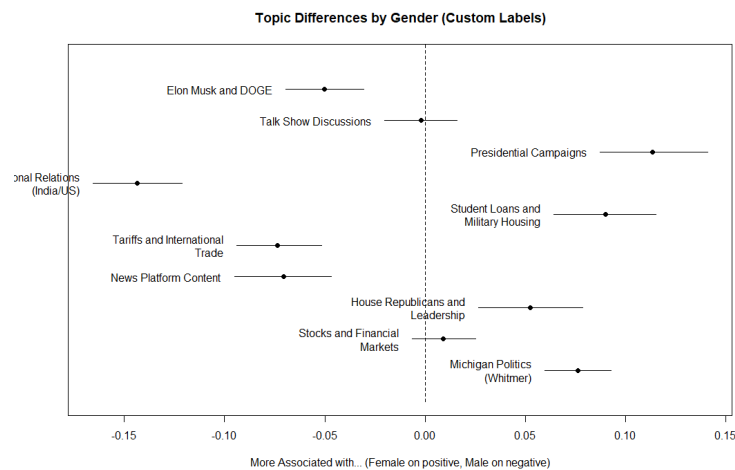
² Each bar represents the expected topic proportion in the corpus, with custom labels assigned for interpretability. For instance, topics like “Presidential Campaigns,” “Trade and Tariffs,” and “News Platform Content” were among the prominent themes uncovered.

The STM with a gender covariate provides estimates of how much more or less prevalent each topic is in female-politician articles compared to male-politician articles. I found that most topics appear in both sets of articles but with varying emphasis.

The topic prevalence analysis revealed a clear gender-based differentiation in the policy focus of news coverage. Topics more associated with female politicians predominantly reflected domestic policy areas. For example, coverage around Presidential Campaigns, Student Loans and Military Housing, and Michigan Politics was more prevalent in articles discussing female politicians. These topics suggest an emphasis on internal governance, social policy, and state-level political dynamics when reporting on women politicians.

Conversely, topics more frequently appearing in articles about male politicians were markedly skewed toward international affairs and economic issues. Themes such as International Relations (India/US), Tariffs and International Trade, and News Platform Content were more associated with male-focused articles. This pattern may indicate a media tendency to frame male politicians within global policy or macroeconomic contexts, whereas female politicians are more often connected with domestic concerns.

This divergence highlights potential media framing biases, where policy domains covered in news articles correlate with the gender of the political figure being discussed.



*Figure 2: Topic prevalence differences by gender.*³

³ Each point represents a topic, positioned on the x-axis by the difference in expected proportion between female-authored and male-authored articles (with 0 meaning no difference). Points to the right (positive values) indicate topics more prevalent in female-politician coverage, while points to the left (negative) indicate topics more prevalent in male-politician coverage.

Emotion Analysis using NRC Lexicon

Beyond topic prevalence, I analyzed the **emotional tone** of the coverage. I utilized the *NRC Word-Emotion Association Lexicon* (often referred to as the NRC Emotion Lexicon) to quantify emotion words in each article. The NRC lexicon is a crowd-sourced lexicon of approximately 14,000 English words manually labeled with one or more of eight basic emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, trust) as well as positive or negative sentiment. I counted how many words from each emotion category appeared in each article. To make fair comparisons, these counts were normalized by article length (per 1,000 words), given that articles can vary in length.

After computing emotion word frequencies for each article, I aggregated the results by gender. For each emotion category, I calculated the mean frequency in female-politician articles and in male-politician articles. The differences were insightful. The figure below shows the average frequency of words for each emotion, comparing female vs. male article sets:

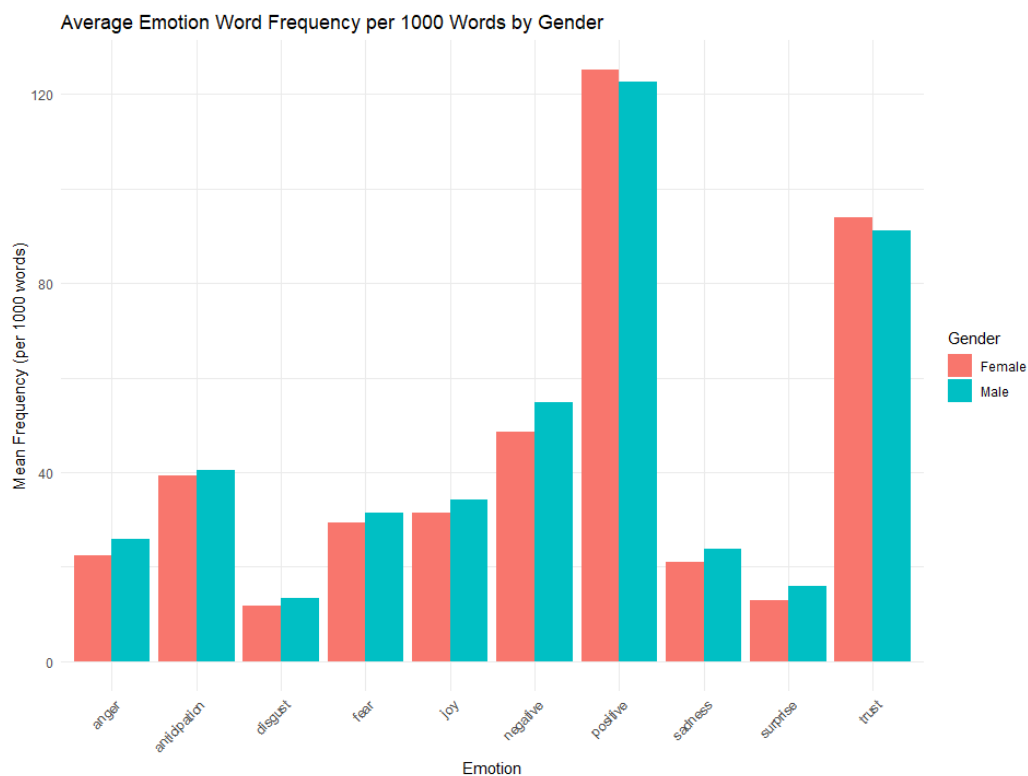


Figure 3: Average emotion word frequency (per 1,000 words) by gender.⁴

I observe that for most emotions, the frequencies in female vs. male coverage are

⁴ Red bars represent articles about female politicians, and teal bars represent articles about male politicians. Emotions on the x-axis include the basic emotions (anger, anticipation, etc.) and overall positive/negative sentiment. For example, words associated with trust are the most common emotion words in both sets of articles (as many political news articles contain words like “said,” “stated,” “confirmed” which are categorized under trust), whereas disgust words are among the least frequent.

quite similar. Both sets of articles contain a mix of positive and negative emotion-laden words, with **positive** and **trust** words being more frequent than very negative emotions. However, when focusing on the **negative emotions** – particularly **anger**, **disgust**, **fear**, and **sadness** – a consistent pattern emerged: these negative emotion words are slightly more frequent in articles about male politicians than those about female politicians. For instance, words conveying anger (e.g., “angry,” “furious”) and disgust (e.g., “corrupt,” “disgusted”) appeared a bit more in the male-focused subset. The differences, while not large in magnitude (often on the order of a few words per thousand), are statistically significant according to t-tests. I performed two-sample t-tests for each of the four negative emotion categories, testing the null hypothesis that there is no difference in mean frequency between female-article and male-article groups. The results showed significantly higher usage of **anger** words in male coverage ($p < 0.001$), higher **disgust** word usage in male coverage ($p < 0.001$), higher **sadness** and **fear** word usage as well (both $p < 0.05$ with sadness $p < 0.001$). This suggests that news articles about male politicians tend to use slightly harsher or more negative emotional language on average.

One possible explanation could be that male politicians are more frequently subjects of scandals or conflict-driven news (which inherently carries negative tone), whereas female politicians might receive more neutral or positive feature coverage. Alternatively, it could reflect a subtle bias where writers temper negative language when writing about female figures. On the positive side, categories like **joy**, **trust**, and **anticipation** did not show significant differences – meaning both female and male-related articles conveyed positive sentiments and forward-looking anticipation at comparable rates. Overall, the emotion analysis provides a quantitative backbone to the notion that *tone* differs by gender in political news. It is important to note that these are aggregate trends and individual articles can vary widely; nevertheless, across 1,050+ articles per group, these differences are meaningful.

Word Embedding Analysis (Semantic Shift)

While topic modeling and lexicons give us high-level differences, I also investigated *semantic differences* – that is, how certain words might be used in different contexts depending on the politician’s gender. For this, I trained **Word2Vec** models on the two subsets of articles. Word2Vec is a neural embedding model that learns vector representations of words such that words frequently appearing in similar contexts have similar vectors. I trained one Word2Vec model on all articles about female politicians and another on all articles about male politicians (using the same settings for both, e.g., continuous bag-of-words architecture, 100-dimensional vectors, 5-word window, 5 training iterations). The separate models allow us to capture the “semantic space” of words in female-focused versus male-focused corpora.

After training, I extracted the vector for each word from both models (restricting to words that appear in both corpora for a fair comparison). I then computed the **cosine similarity** between the female-context vector and the male-context vector for each word. Cosine similarity is a measure from 0 to 1 that indicates how similar the two vector representations are (1 means identical context, 0 means completely different). Words with high cosine similarity have similar meanings/usage in both sets of articles, whereas words with lower cosine similarity may be used in notably different ways.

I ranked words by increasing cosine similarity to find those with the greatest semantic shift between the two contexts. The following table illustrates a subset of words with the lowest cosine similarity, which means most divergent meanings:

Table1: <i>Table of words with the lowest cosine similarity</i>	
	navy
	period
	woman
	age
	track
	tension
	jurisprudence
	round
	usa
	allowed

To make these shifts concrete, I created a **context comparison table** for some of the most shifted words. For each word, I looked at example phrases from the male articles and from the female articles, and I summarized the differing contexts:

Table 2: Context comparison for selected words with semantic shifts

<i>Word</i>	<i>Summary</i>	<i>Example (Male)</i>	<i>Example (Female)</i>
<i>navy</i>	Male: army-navy games and military; Female: navy roles in political discussions and veterans.	army navy game on saturday army and navy met at northwest stadium	department of defense and the u s navy pelosi may believe amazon founder jeff bezos ...
<i>period</i>	Male: historical events; Female: recovery after childbirth.	for a period of rapid change in the energy industry	the period of recovery following the birth
<i>woman</i>	Male: 'first woman' achievements; Female: empowerment and identity.	the first woman to lead the defense department	as a woman of color she brings a new perspective to the senate
<i>age</i>	Male: election eligibility; Female: healthcare and maternal age.	he will reach the minimum age to qualify for the election	maternal age is a significant factor in childbirth outcomes
<i>track</i>	Male: campaign tracking; Female: professional achievements.	tracking the progress of the presidential campaign through each state	her track record in healthcare reform is impressive
<i>tension</i>	Male: diplomatic conflict; Female: rights and healthcare tension.	tension between the us and china over trade policies escalates	tension grew over education funding cuts impacting children
<i>jurisprudence</i>	Male: general legal background; Female: reproductive rights jurisprudence.	jurisprudence shaped his approach to criminal justice reform	her focus on reproductive rights aligns with constitutional jurisprudence
<i>round</i>	Male: election rounds; Female: legislative review rounds.	he advanced to the next round in the senate elections	this bill passed the first round of legislative review
<i>usa</i>	Male: USA in global diplomacy; Female: USA in domestic rights discussions.	the usa imposed new tariffs during diplomatic talks	the usa must continue its commitment to civil rights
<i>allowed</i>	Male: military permissions; Female: civil rights and healthcare access.	soldiers were allowed to defend the border with force if necessary	women were finally allowed access to critical healthcare services

From the context analysis, I interpret that certain terms carry gendered nuances:

- **“Woman”** – In articles about male politicians, the term often appears in a historical phrase (like highlighting the first woman in a role, since the article might mention a female trailblazer in contrast to the male politician). In female-politician articles, “woman” more directly pertains to the subject or women’s issues, indicating empowerment or identity (e.g., describing the politician as a woman and what that represents).
- **“Age”** – In male context, age is mentioned in formal criteria or political timeline contexts (“the candidate will reach the minimum age for office by...”). In female context, age comes up in personal or health contexts, such as childbirth or life stage. This could be because female politicians’ coverage might include aspects of their personal life or health in ways not seen for men.
- **“Tension”** – In male-article contexts, “tension” often refers to geopolitical or diplomatic tensions (e.g., between countries in foreign policy news). In female contexts, “tension” might refer to domestic or social issues, like tension over education funding, as seen in the example, suggesting that the women in our sample were involved in different types of storylines (education, social policy) compared to men (international conflict).
- **“Jurisprudence”** – This word highlights a stark difference: male-context usage was general (legal philosophy background), but female-context usage tied to *reproductive rights jurisprudence* (likely because one of the female politicians was often discussed in context of abortion or women’s rights legal battles). This indicates how female politicians might be linked to women’s issues in coverage.
- Other words like **“period,” “track,” “allowed,” “navy,”** etc., showed similar shifts – often a mundane or general meaning in one context and a gender-specific or unexpected meaning in the other.

This word embedding analysis reveals subtle biases: even when the same terms are used, the subjects and framing can differ by gender. It’s an important reminder that vocabulary alone doesn’t tell the full story – context does. By comparing embeddings, I uncover that female-focused articles can have different thematic associations for words than male-focused ones. These differences likely arise from the distinct narratives surrounding female versus male public figures.

Discussion of Findings

Our analysis demonstrates that there are indeed gender-based differences in political news coverage at multiple levels: topical focus, emotional tone, and even word meaning. Topic modeling revealed that although female and male politicians are covered across a wide spectrum of issues, the thematic emphasis differs systematically. Articles about female politicians were more likely to focus on domestic policy issues, such as presidential campaigns, student loans, military housing, and state-level governance. In contrast, coverage of male politicians leaned toward international relations, trade policies, and macroeconomic topics. This pattern suggests that media narratives may unconsciously assign women politicians to domestic or social policy domains, while associating male politicians with foreign policy and economic leadership.

Interestingly, the traditional assumption that women are disproportionately subjected to “soft news” or stories about their personal lives was not strongly supported in the current dataset. Instead, both male and female politicians were predominantly covered in substantive policy contexts, although the type of policy focus varied by gender.

The emotion analysis added another dimension: media coverage of male politicians contained more frequent negative emotion words (anger, fear, etc.). This could imply that stories about men are more often conflict-oriented or involve criticism, for example, discussions of controversies, attacks, or failures that naturally contain negative language. Female politicians may enjoy slightly more positive or neutral press in terms of language tone, or it could be that criticism of women is delivered in a more nuanced way without overt “anger” words. Another interpretation is that male politicians, often holding or running for higher offices, like presidency, are under heavier scrutiny, thus attracting more negative reportage.

From the word embedding results, I see evidence of what could be called *semantic bias*. The same word can paint different pictures: when reading about a female politician, encountering the word “children” might literally refer to her children or family policy, whereas for a male politician “children” might appear in a policy context, like education policy, rather than about his own kids. These subtle shifts underscore how even neutral terms can contribute to a gendered narrative.

There are several important considerations in interpreting these findings. First, our dataset, while balanced in gender, is not a random sample of all political news – it specifically focuses on 14 well-known figures. The particular issues surrounding those individuals during the sampled time frame could drive some differences. For example,

if one female politician was heavily involved in a reproductive rights debate, that could increase emotion words like “fear” or unique word usages like “jurisprudence” in her articles, not because she is female per se but because of the topic she was engaged in. I tried to mitigate such confounds by the diversity of the sample (multiple politicians), but residual effects can remain. Second, the articles are about politicians, not authored by them. Thus, when I observe differences, it is a product of journalistic choices and possibly public narratives, not the politicians’ own word choices. For instance, the press might more readily highlight the personal lives of female politicians. Third, using LexisNexis means I likely have more traditional and established news outlets represented; the findings might differ if social media or less formal news were included. Also, our emotion analysis relies on a predefined lexicon which may not catch sarcasm or context – though it’s a standard tool, it has limitations in understanding sentiment in context.

Despite these caveats, the convergence of evidence across methods strengthens our confidence that the observed gender differences are real. Where one method might be noisy, the combination (topics, lexicon sentiments, embedding) provides cross-validation.

Conclusion and Future Directions

Female politicians are more often associated with domestic policy topics, such as social policy, education, and state-level governance, while male politicians receive relatively more coverage on international affairs, trade, and macroeconomic policy. Emotionally, articles about female politicians tend to use slightly gentler language, with lower frequencies of negative emotional words such as anger, disgust, fear, and sadness. In contrast, coverage of male politicians contains a marginally higher prevalence of such negative emotions, suggesting a somewhat harsher or more confrontational tone.

Although the magnitude of these differences is modest, the patterns are consistent across multiple analytical dimensions – topics, emotional tone, and semantic usage. This subtle framing could cumulatively influence public perceptions. For instance, voters might unconsciously associate female candidates with domestic or personality-oriented traits, while linking male candidates with leadership in foreign policy or economic domains. Over time, such media portrayals may reinforce gender stereotypes that shape both public attitudes and electoral outcomes.

Moving forward, there are several avenues to extend this research. A natural next step is to expand the dataset to more politicians (including less prominent ones) and additional news sources, including international media, to see if these patterns hold broadly. It would also be valuable to analyze temporal aspects: has the gap in topics or

tone narrowed in recent years as more women enter politics, or does it spike during election campaigns? Another direction is a more fine-grained linguistic analysis using advanced language models (e.g., BERT-based contextual embeddings) to detect bias in phrasing that the current lexicon or static embedding approach might miss. I could also incorporate author-level analysis. Are articles about female politicians more often written by female journalists, and does that affect the tone? Finally, exploring readers' reception through surveys or experiments of differently framed articles could connect these textual findings to real-world impact.

In conclusion, by combining STM, emotion lexicons, and word embeddings, our analysis provides a comprehensive view of gendered differences in political news coverage. This multi-method approach serves as a template for analyzing media bias in other contexts as well. I hope our findings will contribute to ongoing discussions about media representation and encourage news professionals to be mindful of implicit biases in their reporting. The ultimate goal is a political news landscape where coverage is fair, balanced, and free of gendered double standards, focusing on each politician's actions and qualifications above all else.